

# SOC Investigation Report — Suspicious HTTP Beaconing Investigation

| Field                       | Value                                                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case ID                     | SOC-LAB-2026-0805-BEACON                                                                                                                                                                      |
| Classification              | Detection Engineering Lab / Behavioral Threat Hunting Exercise                                                                                                                                |
| Host                        | WIN11-CLIENT01                                                                                                                                                                                |
| Process Under Investigation | InventoryAgent.exe (Windows Service)                                                                                                                                                          |
| Service Name                | InventoryAgent                                                                                                                                                                                |
| Data Source                 | Sysmon (Operational log) + Windows System log → Splunk (index=sysmon)                                                                                                                         |
| Date of Activity            | 2026-08-05 (03:49:06 – 04:08:21 UTC, ongoing at time of capture)                                                                                                                              |
| Alert Triggered             | Periodic Outbound HTTPS from SYSTEM-Level Service — Possible Beaconing                                                                                                                        |
| Initial Severity            | Medium — behaviorally consistent with beaconing, but no compromise indicators (no injection, no PowerShell, no malicious persistence, no child processes, no credential access) were observed |
| Outcome                     | Resolved — Legitimate Software (not provable from Sysmon telemetry alone; required follow-up validation)                                                                                      |

## 1. Objective

Investigate a long-running Windows service exhibiting periodic, encrypted outbound connections to determine whether the behavior represents authorized enterprise software or command-and-control (C2) beaconing. This report deliberately avoids the label "C2 Detection" — no attacker tasking, payload delivery, or malicious execution was observed. What was observed is a behavioral pattern (service-based persistence, SYSTEM execution, regular encrypted check-ins to a fixed destination) that is common to both legitimate management agents and real beacon malware, and the investigation reflects that ambiguity honestly rather than assuming intent either way.

## 2. Investigation Scenario & Verdict

A process named `InventoryAgent.exe`, running as a Windows Service under NT AUTHORITY\SYSTEM, was observed starting automatically at boot, resolving a fixed internal hostname, and then establishing an HTTPS connection to that host roughly once per minute, indefinitely.

**Verdict: Confirmed legitimate after validation against application ownership and implementation details.** Subsequent validation against asset ownership records and the known implementation of this internally developed inventory service confirmed the process performs only host registration and periodic status "heartbeats," with no command execution, file operations, or data exfiltration capability. Critically, **this verdict was only reachable through that follow-up validation — not from Sysmon telemetry alone.**

The investigation below is written as it would be conducted before that validation occurred, since that is the realistic starting position for a SOC analyst encountering this pattern on an unfamiliar host.

### 3. Initial Detection

```
Query: index=sysmon EventCode=1 Image="*\\InventoryAgent.exe"
```

#### Finding

`InventoryAgent.exe` (PID 3524, ProcessGuid {0875a023-b2db-6a72-4500-000000006100}) started at 03:49:47.378 UTC, parented directly by `services.exe` (PID 964) rather than any interactive shell or user session, running at **System** integrity under **NT AUTHORITY\SYSTEM**.

### 4. Initial Assessment — Why the Activity Was Flagged

| Indicator                                       | Detail                                                                                                                                                      |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service-based execution                         | Started by <code>services.exe</code> , not a user session — indicates a registered Windows Service with automatic startup, not one-off interactive activity |
| SYSTEM-level execution                          | Maximum local privilege; also the norm for legitimate endpoint agents (EDR, patch management, RMM tools), so this alone is not distinguishing               |
| DNS resolution immediately preceding connection | The process resolved a specific hostname just before its first outbound connection — deliberate, configured behavior rather than incidental                 |
| Regular outbound cadence                        | Repeated HTTPS connections to the same destination at roughly one-minute intervals, sustained for the full observation window                               |
| Fixed destination, stable ProcessGuid           | Every connection traces to the same process instance and the same destination — a single long-lived channel, not scattered one-off traffic                  |

None of these indicators, individually or together, distinguish legitimate management software from a C2 beacon — that determination requires evidence outside Sysmon telemetry (Section 10).

### 5. Process & Persistence Investigation

```
Query: index=sysmon EventCode=1 ProcessGuid="{0875a023-b2db-6a72-4500-000000006100}"
```

#### Finding

| Field  | Value                                              |
|--------|----------------------------------------------------|
| Image  | C:\Program Files\InventoryAgent\InventoryAgent.exe |
| Parent | services.exe (PID 964)                             |
| User   | NT AUTHORITY\SYSTEM                                |

| Field           | Value                                                            |
|-----------------|------------------------------------------------------------------|
| Integrity Level | System                                                           |
| LogonId         | 0x3E7 (the standard SYSTEM logon session)                        |
| SHA256          | 14945B284FD3E23F08F8CBE59F7C214453F4BDDC31791D46E0FEA6230D0C39F8 |

The process started 41 seconds after the host itself booted (Windows Event ID 12, 2026-08-05T03:49:06.500Z), consistent with a service configured for automatic startup rather than delayed or manually triggered start.

The service automatically started after system boot without any user interaction. This is an important behavioral characteristic in its own right: it means the process is configured for persistence at the service level, independent of any user logging on — the same behavior legitimate management agents and malicious persistence mechanisms both rely on.

**Analyst note — PE metadata artifact:** the binary's `OriginalFileName` field reads `InventoryAgent.dll`, not `.exe` — the same benign .NET single-file-publish artifact observed in the BITS investigation in this series, not evidence of tampering. Similar metadata inconsistencies are common in .NET single-file published applications and should not be interpreted as evidence of masquerading without additional supporting evidence

## 6. DNS Investigation

```
Query: index=sysmon EventCode=22 ProcessGuid="{0875a023-b2db-6a72-4500-000000006100}"
```

### Finding

Two DNS queries were issued by the process within the same second, immediately preceding its first network connection:

| Time (UTC)   | Query                 | Result                    | Purpose                                                                                 |
|--------------|-----------------------|---------------------------|-----------------------------------------------------------------------------------------|
| 03:51:19.307 | WIN11-CLIENT01 (self) | Local IPv4/IPv6 addresses | Resolving its own hostname to enumerate local IP addresses for its registration payload |
| 03:51:19.485 | inventory.corp.local  | 192.168.56.107            | Resolving its configured remote server before connecting                                |

The self-resolution of `WIN11-CLIENT01` is consistent with the application collecting local network information before registration, rather than any reconnaissance or discovery activity.

## 7. Network Activity & Beacon Interval Analysis

```
Query: index=sysmon EventCode=3 ProcessGuid="{0875a023-b2db-6a72-4500-000000006100}"
```

## Finding

The initial connection corresponds to application registration, followed by recurring heartbeat traffic to the same destination. The first connection (03:51:19.719 UTC, ~92 seconds after process start) established a TCP session to 192.168.56.107:8443 (inventory.corp.local); 17 further connections to the same destination followed through the end of the capture window (04:08:21.266 UTC), with the process still running and presumably continuing beyond it.

**Interval statistics** (time between consecutive connections):

| Metric           | Value                                                                            |
|------------------|----------------------------------------------------------------------------------|
| Average interval | 60.0 seconds                                                                     |
| Minimum interval | 38.1 seconds                                                                     |
| Maximum interval | 81.1 seconds                                                                     |
| Destination      | inventory.corp.local (192.168.56.107:8443), consistent across all 18 connections |

The regularity here (tight clustering around a 60-second mean) is itself an artifact worth noting: genuinely human-driven traffic doesn't look like this, while both legitimate scheduled agents and beacon malware commonly do. The presence of jitter (a 43-second spread rather than a fixed interval) is a detail attackers and legitimate developers alike build in deliberately, to avoid the flat, perfectly-periodic pattern that naive interval-based detection rules look for.

## 8. Activity Chain



## 9. Timeline Reconstruction (UTC)

| Time                  | Activity                                                 |
|-----------------------|----------------------------------------------------------|
| 03:49:06.500          | Host boots (Windows Event ID 12)                         |
| 03:49:47.378          | InventoryAgent.exe started by services.exe               |
| 03:51:19.307 – 19.485 | DNS resolution: self-hostname, then inventory.corp.local |

| Time                        | Activity                                                                                                |
|-----------------------------|---------------------------------------------------------------------------------------------------------|
| 03:51:19.719                | First HTTPS connection established (registration)                                                       |
| 03:51:19.719 – 04:08:21.266 | 17 additional HTTPS connections — recurring heartbeat traffic — at ~60s average interval (range 38–81s) |

Total observed window: ~**19 minutes**, with the service still active at the end of the capture — this is a sustained, ongoing channel, not a bounded event.

## 10. Resolution Criteria — What Would Confirm or Refute Malicious Intent

Sysmon telemetry alone cannot resolve this case. A real investigation would additionally require:

| Check                                                               | What It Would Tell You                                                                                                     |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Code signing / digital signature on <code>InventoryAgent.exe</code> | Legitimate enterprise agents are almost always signed; an unsigned binary running as SYSTEM is a strong escalation trigger |
| Asset inventory / CMDB cross-reference                              | Confirms whether this software was ever knowingly deployed to this host by IT/security                                     |
| TLS certificate presented by <code>inventory.corp.local:8443</code> | A valid internal PKI certificate supports legitimacy; a self-signed or mismatched certificate is a red flag                |
| DNS zone ownership for <code>corp.local</code>                      | Confirms whether the destination is genuinely internal infrastructure or an attacker-controlled name mimicking one         |
| Payload content (via TLS inspection/proxy, if available)            | Directly distinguishes an inventory JSON payload from C2 tasking data                                                      |
| EDR/AV verdict on the binary hash                                   | Independent confirmation from a source other than Sysmon                                                                   |

**Validation finding relevant to this checklist:** follow-up validation of the internal application's known implementation confirmed it disables TLS certificate validation entirely (accepting any server certificate without verification), a pattern typically justified for tolerating a self-signed certificate in an internal environment. This is worth flagging independent of this specific case: **beacon malware very commonly disables certificate validation for the same reason** — to tolerate a self-signed certificate on the far end without a handshake failure. This detail would not be visible from Sysmon alone; it would only surface through binary/traffic analysis (e.g., a TLS inspection proxy or a JA3/JA3S fingerprint that flags non-standard TLS stack behavior).

## 11. MITRE ATT&CK Mapping

| Tactic              | Technique                                        | ID        | Evidence                                                                                                                                                               |
|---------------------|--------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Persistence         | Create or Modify System Process: Windows Service | T1543.003 | Demonstrated as a persistence mechanism in the lab: <code>InventoryAgent.exe</code> registered as an auto-start Windows Service, launched by <code>services.exe</code> |
| Command and Control | Application Layer Protocol: Web Protocols        | T1071.001 | Periodic HTTPS connections to a fixed destination at regular intervals                                                                                                 |

## 12. Indicators of Interest

*(Named "Indicators of Interest" rather than "Indicators of Compromise" — compromise was not established; these are the artifacts that would carry over into a real investigation of this pattern.)*

| Type              | Value                                                            |
|-------------------|------------------------------------------------------------------|
| Process           | InventoryAgent.exe                                               |
| Service Name      | InventoryAgent                                                   |
| SHA256            | 14945B284FD3E23F08F8CBE59F7C214453F4BDDC31791D46E0FEA6230D0C39F8 |
| Parent Process    | services.exe                                                     |
| Execution Context | NT AUTHORITY\SYSTEM                                              |
| Destination Host  | inventory.corp.local                                             |
| Destination IP    | 192.168.56.107                                                   |
| Destination Port  | 8443 (HTTPS)                                                     |
| Beacon Interval   | ~60s average (38–81s range)                                      |
| ProcessGuid       | {0875a023-b2db-6a72-4500-000000006100}                           |

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## 13. Threat Assessment

This behavioral pattern — SYSTEM-level service, automatic startup, DNS resolution followed by regular encrypted check-ins to a single destination — is functionally identical to how real C2 beacon frameworks operate, and is equally how legitimate management agents (CrowdStrike, Tanium, SentinelOne, RMM tools) behave. That overlap is the point: signature and hash-based detection cannot address this pattern, since a renamed or newly-compiled beacon would produce identical Sysmon telemetry to this lab's benign agent. Effective detection here depends on layered context — asset ownership records, code signing, TLS certificate validation, and destination reputation — rather than any single Sysmon event.

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## 14. Splunk Threat Hunting & Correlation Queries

### Identify Long-Running Processes with Regular Outbound Connections

```
index=sysmon EventCode=3
| stats count earliest(_time) as first_seen latest(_time) as last_seen by
Computer Image DestinationHostname DestinationPort ProcessGuid
| where count > 5
| eval duration_min=round((last_seen-first_seen)/60,1)
| table Computer Image DestinationHostname DestinationPort count duration_min
```

*Purpose:* Surfaces any process making repeated connections to the same destination over a sustained window — the general-purpose version of this investigation, not scoped to a known process name.

## Beacon Interval Analysis

```
index=sysmon EventCode=3 Image="*\\InventoryAgent.exe"  
| sort _time  
| streamstats current=f last(_time) as PreviousTime  
| eval Interval=round(_time-PreviousTime,1)  
| table _time Interval DestinationHostname DestinationPort
```

*Purpose:* Quantifies the interval between connections directly, rather than relying on eyeballing timestamps — this produced the 60s average / 38–81s range cited in Section 7.

## Connection Frequency Over Time

```
index=sysmon EventCode=3 Image="*\\InventoryAgent.exe"  
| timechart span=1m count
```

*Purpose:* Visualizes the steady cadence at a glance — a flat, near-constant rate per minute is itself the signal worth investigating.

## Correlate Full Process Lineage

```
index=sysmon (ProcessGuid="{0875a023-b2db-6a72-4500-000000006100}" OR  
ParentProcessGuid="{0875a023-b2db-6a72-4500-000000006100}")  
| sort _time  
| table _time EventCode Image ParentImage CommandLine
```

*Purpose:* Confirms the process spawned no children during the observation window — relevant because a beacon that receives tasking would typically be expected to eventually spawn a child process or write a file, neither of which occurred here.

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# 15. Detection Engineering

**Monitoring coverage** — this behavioral pattern is detectable by alerting on:

- New services registered with SYSTEM-level auto-start that were not present in a prior known-good baseline
- Processes generating more than N outbound connections to the same destination within a rolling window, regardless of process name
- Statistical regularity in connection timing (low variance in inter-connection interval) as a standalone signal, independent of destination reputation
- Unsigned or unknown binaries running as SYSTEM with active outbound network activity

### Actionable rule specs:

1. **Baseline deviation rule:** alert when a newly-registered Windows Service establishes its first outbound network connection within minutes of installation — legitimate software rollouts are typically known in advance via change management, so an unexpected new service phoning home is a meaningful signal on its own.

2. **Beacon-interval rule:** flag processes whose Event ID 3 connections to a single destination show a coefficient of variation below a defined threshold over a rolling 15-connection window — mathematically distinguishing "regular check-in" traffic from ordinary application chatter.
  3. **Enrichment requirement:** any alert on this pattern should auto-attach code-signing status and CMDB/asset-inventory lookup results before reaching an analyst, since — per Section 10 — the Sysmon evidence alone cannot resolve the verdict.
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## 16. Recommended Response Actions

1. **Do not immediately assume compromise** — this pattern alone is insufficient grounds for isolation; escalate to verification first (Section 10 checklist).
  2. **Check code signing and hash reputation** on the binary before further action.
  3. **Cross-reference asset inventory / CMDB** to confirm whether this software was knowingly deployed.
  4. **Inspect the TLS certificate** presented by the destination host, and confirm DNS zone ownership for the resolved domain.
  5. **If unresolved after the above**, escalate to network-level inspection (proxy logs, TLS decryption if available) to examine payload content directly.
  6. **If confirmed malicious**, stop the service (`sc stop <name>`), remove its service registration, and treat the destination as a confirmed C2 indicator for environment-wide hunting.
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## 17. Conclusion

This investigation reconstructed a complete behavioral pattern — service-based SYSTEM execution, DNS resolution, and sustained periodic HTTPS connections averaging 60 seconds apart — using Sysmon Event IDs 1, 3, and 22 correlated through a single stable ProcessGuid. The pattern is textbook beaconing behavior from a telemetry standpoint, and the investigation deliberately does not overstate that into a "C2 detected" finding, because the same telemetry is equally consistent with legitimate management software.

The ground truth in this lab (a validated, internally developed inventory agent) is not something Sysmon telemetry alone could establish — and that is the central lesson of this investigation: **behavioral detection surfaces the right cases for review, but resolving them requires context Sysmon does not provide.** The resolution criteria in Section 10, not the raw telemetry, are what a real SOC investigation of this pattern would actually turn on.